

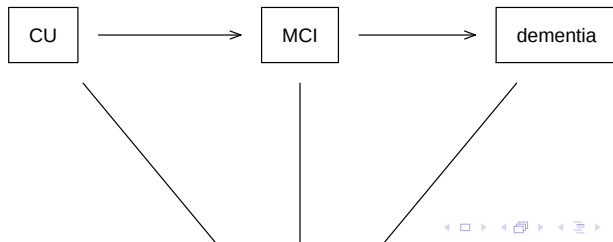
Well, I generally don't generate ideas of my own. Someone has to come to me with a problem. And, apparently, I'm pretty good at coming up with solutions or ideas for solutions. Identifying problems is what is important – solutions just follow. Nathan Mantel

Four basic premises

1. Science first
2. Multi-state models are crucial
3. Both rates *and* absolute risk are needed
4. The perfect one number summary is a search for El Dorado

Multistate models

```
library(survival)
sname <- c("CU", "MCI", "dementia", "death")
smat2 <- matrix(0, 4, 4, dimnames=list(from= sname, to= sname))
smat2[1,2] <- smat2[2,3] <- 1
smat2[1:3,4] <- 1
par(mar=c(1,1,0,1))
statefig(rbind(3,1), smat2)
```



Multi-state models

- ▶ Arrows: rates, hazard ratios
- ▶ Boxes: absolute risk
 1. $p_k(t; x)$: probability in state k at time t
 2. $E[N_k(t; x)]$: number of visits to state k , by time t
 3. $\text{sojourn}_k(t; x)$: total time in state k , up to time t
 4. $\text{sojourn}_k(t; x)/E[N_k(t; z)]$: expected time per visit
 5. $P_{jk}(t; x)$: a matrix, probability will be in state k at time t , given in state j at time s

Form 1: extension of KM and Cox

- ▶ Model the *transitions*
- ▶ Prob(subject i transitions from j to k at time t)
- ▶ Crazy fact: $PLL = \sum_{jk} PLL_{jk}$
- ▶ PLL = standard Cox partial log likelihood
- ▶ survfit does both single state (KM) and multi-state (AJ)
- ▶ coxph does both single state and multi-state
- ▶ Absolute risk requires all the λ_{jk} estimates

Form 2: interval censored multistate

- ▶ Model the *observed state*
- ▶ $P(\text{in observed state } k \text{ at time } t)$
- ▶ Fit all states at once
 - ▶ for a trial value $\hat{\beta}$ of β
 - ▶ compute $\hat{p}(t; x_i, \hat{\beta})$, for each subject
 - ▶ compare $\hat{p}$ to observed state(s), update $\hat{\beta}$
- ▶ Allows for much more general observations
 - ▶ observed state
 - ▶ subset of states
 - ▶ biomarker of state (HMM)

Questions

- ▶ Cost/benefit of IC fits
- ▶ Time-dependent covariates and absolute risk
- ▶ External validation

Tools

- ▶ Studies
- ▶ Data setup
- ▶ Models and code
- ▶ Models and underlying mathematics
 - ▶ simple
 - ▶ computer
 - ▶ formal

Cox partial likelihood

$$PL(\beta) = \prod_{i \in D} \frac{dN_i(t) e^{x_i \beta}}{\sum_j Y_j(t) e^{x_j \beta}}$$

- ▶ lottery model
- ▶ efficient computation $O(n) + O(dp^2)$
- ▶ Appendix A.3 in the book, simple + computer
- ▶ Cook, formal